CAMREN KHOURY

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EDUCATION

Clemson University – Clemson, SC
B.S. Computer Engineering
B.S. Electrical Engineering

Focus: Embedded systems, Intelligent Systems, Firmware, PCB Design

PUBLICATIONS

WildFireVQA: A Large-Scale Radiometric Thermal VQA Benchmark for Aerial Wildfire Monitoring
Mobin Habibpour, Niloufar Alipour, John Spodnik, **Camren J. Khoury**, Fatemeh Afghah
arXiv:2604.20190, DOI: 10.48550/arXiv.2604.20190, April 2026

EXPERIENCE

Clemson University – IS-WiN Lab — Clemson, SC
Undergraduate Research Assistant (FireSense / FlameVQA) — Jan 2026 – May 2026

- Processed and validated **50,000+** multimodal wildfire samples, enabling large-scale evaluation of AI detection systems
- Built automated RGB–thermal pairing pipeline using EXIF/timestamp logic, eliminating manual dataset alignment
- Implemented feature-based image registration using **SIFT** keypoints and **RANSAC** transform estimation for RGB–thermal alignment refinement
- Developed validation and fallback logic for low-feature thermal imagery, improving robustness across wildfire datasets
- Restructured unorganized field data into training-ready formats (CSV + JSON), improving usability across workflows
- Adapted pipeline to support non-standard sensor formats, increasing compatibility across datasets

Athena Consulting Group — North Charleston, SC
Cybersecurity Intern — May 2022 – Aug 2022

- Supported vulnerability scanning and patch compliance, improving enterprise system security posture
- Automated reporting workflows, reducing manual audit time and improving efficiency
- Assisted in system monitoring and remediation documentation

TECHNICAL PROJECTS

Automated Drosophila Sorting System — Embedded + PCB + Vision
Department of Electrical and Computer Engineering — People's Choice Award, 2026

- Developed sorting platform achieving **4 flies/min throughput** with reduced handling and repeatable operation
- Implemented **10-second autonomous cycle**, supporting continuous high-throughput operation across full sorting pipeline
- Designed motion control system enabling reliable 3-D positioning and consistent fly acquisition across gantry workspace
- Integrated computer vision pipeline for real-time detection, classification (male/female), and routing decisions
- Designed custom PCBs consolidating power distribution, motor drivers, and GPIO, reducing wiring complexity and failure points

Remote Water Quality Monitoring System — IoT + Embedded + PCB

- Designing low-cost water monitoring system for underserved regions lacking sanitation infrastructure
- Reduced projected system cost by **60–80%** compared to commercial solutions through custom PCB integration
- Integrated multi-sensor platform (pH, turbidity, conductivity, ORP, temperature) for comprehensive monitoring
- Engineered system for continuous operation in flowing water environments with minimal maintenance
- Developed cloud-connected architecture enabling **real-time** remote monitoring and data access

Raspberry Pi High-Performance Cluster — Linux + Distributed Systems

- Built **4-node** cluster enabling parallel computation across distributed hardware
- Configured MPI and NFS architecture to support scalable workload execution
- Demonstrated performance gains through parallel benchmarking using HPL

PoolShark Automated Billiards System — Embedded + Control + Vision

- Developed embedded actuation system using motors, servos, and solenoids
- Implemented **PID** control improving accuracy and repeatability of mechanical actions
- Applied computer vision techniques for real-time object detection and tracking

TECHNICAL SKILLS

Embedded Systems: ESP32, Raspberry Pi, GPIO, PWM, ADC, UART, I2C, SPI
PCB Design: KiCad, multi-layer routing, power distribution, DRC, Gerber/Drill Files
Circuits: Analog/digital design, MOSFET switching, signal conditioning, op-amps
Control Systems: PID, Simulink, Ladder Logic, PLC
Computer Vision: OpenCV, segmentation, edge detection
Programming: C, Python, MATLAB, SQL, JavaScript
Tools/Data: Linux, Supabase, AWS, Git/GitHub, MPI, NFS, SSH, LTSpice